



Module Presenter's Manual *for* **Creating Services for the Web**

***Effective from: June 2024
Ver. 1.0***



Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	June 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Learn about Jakarta Web Services Framework and Technologies
- Understand Service Oriented Architecture (SOA)
- Learn Jakarta Persistence API JAX-RS
- Learn Jakarta RESTful Web Services (JAX-RS)
- Learn Jakarta Web Services with JAX-WS
- Learn Java Architecture for XML Binding (JAXB)
- Develop Web Services using Jersey

2. Information on Session Allocation

Module	Online (No. of Hours)
Creating Services for the Web	12

Throughout this Presenter's Manual, the module **Creating Services for the Web** will be referred to as **WEBSRV**.

3. Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 WEBSRV- TL1	Session 2 WEBSRV- TL2	Session 3 WEBSRV- TL3	Session 4 WEBSRV- TL4
2	Session 5 WEBSRV- TL5	Session 6 WEBSRV- TL6		

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	WEBSRV-TL1	All the topics as listed below from Session 1 and Session 2 of <i>Jakarta Web Services</i> book should be covered in this session. <u>Session 1 – Introduction to Web Services</u> ➤ Understand Web Services ➤ Describe types of Web Services ➤ Explain SOAP Web Services ➤ Describe RESTful Web Services <u>Session 2 – Introduction to Jakarta Persistence API (JPA)</u> ➤ Describe Jakarta Persistence ➤ Explain Jakarta Persistence Query Language ➤ Describe Object-Relational Metadata ➤ Outline related technologies such as Hibernate and Spring Data JPA ➤ List JPA versions	<u>Jakarta Web Services</u> SG – Session 1 & 2 XP – Session 1 & 2 TG – Session 1 & 2
2	WEBSRV-TL2	All the topics as listed below from Session 3 of <i>Jakarta Web Services</i> book should be covered in this session. <u>Session 3 – Jakarta RESTful Web Services (JAX-RS)</u> ➤ Outline JAX-RS ➤ Explain JAX-RS ➤ Explain the uses of JAX-RS ➤ Distinguish between JAX-RS and JAX-WS	<u>Jakarta Web Services</u> SG – Session 3 XP – Session 3 TG – Session 3
3	WEBSRV-TL3	All the topics as listed below from Session 3 and Session 4 of <i>Jakarta</i>	<u>Jakarta Web Services</u>

Session No.	Session Title	Session Details	Deliverables' Mapping
		<i>Web Services</i> book should be covered in this session. <u>Session 4 – Jakarta Web Services with JAX-WS (JPA)</u> ➤ Describe JAX-WS ➤ Explain SOAP Web Services using JAX-WS ➤ Identify types supported by JAX-WS ➤ Describe client support for JAX-WS ➤ Explain Web service interoperability in JAX-WS ➤ Explain generating the Stubs	SG – Session 4 XP – Session 4 TG – Session 4
4	WEBSRV-TL4	The Try It Yourself questions from Session 1 to Session 4 of <i>Jakarta Web Services</i> book should be covered in this session.	<u>Jakarta Web Services</u> Session 1-4
5	WEBSRV-TL5	All the topics as listed below of Session 5 of <i>Jakarta Web Services</i> book should be covered in this session. <u>Session 5 – Java Architecture for XML Binding (JAXB)</u> ➤ Explain JAXB ➤ Describe marshalling and unmarshalling in JAXB ➤ Explain the importance of JAXB for XML schemas ➤ Discuss best practices for XML/Schema Data Binding ➤ Distinguish between JAX-RS, JAX-WS, and JAXB ➤ Identify advantages and disadvantages of JAX-RS, JAX-WS, and JAXB <u>Session 6 – Mini Project: Example of Developing Web Services Using Jersey</u>	<u>Jakarta Web Services</u> SG – Session 5 XP – Session 5 TG – Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain Jersey RESTful Web services ➤ Explain Gradle ➤ Describe the implementation of Jersey in Gradle-managed project ➤ Explain creating different HTTP requests in Resource class	
6	WEBSRV-TL6	The Try It Yourself questions from Session 5 and Session 6 of <i>Jakarta Web Services</i> book should be covered in this session.	<u>Jakarta Web Services</u> Session 5 & 6

6. Library References

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| ➤ Jakarta EE Cookbook Second Edition by Elder Moraes |
| ➤ Jakarta EE for Java Developers: Build Cloud-Native and Enterprise Applications Using a High-Performance Enterprise Java Platform by Rhuan Rocha |

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